

Ex.no:3	DATA CLEANING

Aim:

To identify and handle missing values in the City Day Air Quality dataset using Pandas by applying different data cleaning techniques such as filling missing values, forward fill, backward fill, and dropping missing records.

Algorithm:

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv("city_day.csv")

# Check missing values
print("===== DATASET INFO =====")
```

```

df.info()

print("\n===== MISSING VALUES =====")

print(df.isnull().sum())

print("\n===== TOTAL MISSING VALUES =====")

print(df.isnull().sum().sum())

print("\n===== ANY MISSING VALUES? =====")

print(df.isnull().values.any())

print("\n===== NON-MISSING VALUES COUNT =====")

print(df.notnull().sum())

# Fill missing values with constant
df_constant = df.fillna(0)

print("\n===== CONSTANT VALUE =====")

print(df_constant.head())

# Fill AQI with mean
df_mean = df.copy()

df_mean["AQI"] = df_mean["AQI"].fillna(df_mean["AQI"].mean())

print("\n===== MEAN =====")

print(df_mean[["City", "Date", "AQI"]].head())

# Fill AQI with median
df_median = df.copy()

df_median["AQI"] = df_median["AQI"].fillna(df_median["AQI"].median())

print("\n===== MEDIAN =====")

print(df_median[["City", "Date", "AQI"]].head())# Fill AQI_Bucket with mode
df_mode = df.copy()

df_mode["AQI_Bucket"] = df_mode["AQI_Bucket"].fillna(
    df_mode["AQI_Bucket"].mode()[0]
)

print("\n===== MODE =====")

print(df_mode[["City", "Date", "AQI_Bucket"]].head())

# Forward Fill

```

```
df_ffill = df.ffmpeg()
print("\n===== FORWARD FILL =====")
print(df_ffill.head())
# Backward Fill
df_bfill = df.bfill()
print("\n===== BACKWARD FILL =====")
print(df_bfill.head())
# Drop rows with missing values
df_drop_rows = df.dropna()
print("\n===== DROP ROWS =====")
print(df_drop_rows.head())
# Drop columns with missing values
df_drop_cols = df.dropna(axis=1)
print("\n===== DROP COLUMNS =====")
print(df_drop_cols.head())
# Drop rows having any missing value
df_any = df.dropna(how="any")
print("\n===== DROP HOW='ANY' =====")
print(df_any.head())
# Drop rows where all values are missing
df_all = df.dropna(how="all")
print("\n===== DROP HOW='ALL' =====")
print(df_all.head())
# Keep rows with at least 10 non-null values
df_thresh = df.dropna(thresh=10)
print("\n===== THRESH=10 =====")
print(df_thresh.head())
# Drop rows where AQI or AQI_Bucket is missing
df_subset = df.dropna(subset=["AQI", "AQI_Bucket"])
print("\n===== SUBSET (AQI, AQI_Bucket) =====")
```

```

print(df_subset.head())

# Final cleaned dataset

df_clean = df.copy()

num_cols = df_clean.select_dtypes(include=np.number).columns

df_clean[num_cols] = df_clean[num_cols].fillna(
    df_clean[num_cols].median()
)

cat_cols = df_clean.select_dtypes(include="object").columns

for col in cat_cols:
    df_clean[col] = df_clean[col].fillna(
        df_clean[col].mode()[0]
    )

print("\n===== FINAL CLEANED DATASET =====")

print(df_clean.head())

print("\n===== MISSING VALUES AFTER CLEANING =====")

print(df_clean.isnull().sum())

print("\nData cleaning completed successfully.")

```

Output:

```

===== DATASET INFO =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   City                  29531 non-null  object
 1   Date                  29531 non-null  object
 2   PM2.5                 24933 non-null  float64
 3   PM10                  18391 non-null  float64
 4   NO                     25949 non-null  float64
 5   NO2                   25946 non-null  float64
 6   NOX                   25346 non-null  float64
 7   NH3                   19203 non-null  float64
 8   CO                    27472 non-null  float64
 9   SO2                   25677 non-null  float64
10   O3                    25509 non-null  float64
11   Benzene               23908 non-null  float64
12   Toluene               21490 non-null  float64
13   Xylene                11422 non-null  float64
14   AQI                   24850 non-null  float64
15   AQI_Bucket            24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB

===== MISSING VALUES =====
City                0
Date                0

```

===== MISSING VALUES =====

```
City          0
Date          0
PM2.5        4598
PM10         11140
NO           3582
NO2          3585
NOx          4185
NH3         10328
CO           2059
SO2          3854
O3           4022
Benzene      5623
Toluene     8041
Xylene      18109
AQI          4681
AQI_Bucket   4681
dtype: int64
```

===== TOTAL MISSING VALUES =====

88488

===== ANY MISSING VALUES? =====

True

===== NON-MISSING VALUES COUNT =====

```
City          29531
Date          29531
PM2.5        24933
PM10         18391
NO           25949
NO2          25946
NOx          25346
NH3         19203
CO           27472
SO2          25677
O3           25509
Benzene      23908
Toluene     21490
Xylene      11422
AQI          24850
AQI_Bucket   24850
dtype: int64
```

===== CONSTANT VALUE =====

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	\
0	Ahmedabad	2015-01-01	0.0	0.0	0.92	18.22	17.15	0.0	0.92	27.64	
1	Ahmedabad	2015-01-02	0.0	0.0	0.97	15.69	16.46	0.0	0.97	24.55	
2	Ahmedabad	2015-01-03	0.0	0.0	17.40	19.30	29.70	0.0	17.40	29.07	
3	Ahmedabad	2015-01-04	0.0	0.0	1.70	18.48	17.97	0.0	1.70	18.59	
4	Ahmedabad	2015-01-05	0.0	0.0	22.10	21.42	37.76	0.0	22.10	39.33	

	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	133.36	0.00	0.02	0.00	0.0	0

```

===== MEAN =====
City      Date      AQI
0 Ahmedabad 2015-01-01 166.463581
1 Ahmedabad 2015-01-02 166.463581
2 Ahmedabad 2015-01-03 166.463581
3 Ahmedabad 2015-01-04 166.463581
4 Ahmedabad 2015-01-05 166.463581

===== MEDIAN =====
City      Date      AQI
0 Ahmedabad 2015-01-01 118.0
1 Ahmedabad 2015-01-02 118.0
2 Ahmedabad 2015-01-03 118.0
3 Ahmedabad 2015-01-04 118.0
4 Ahmedabad 2015-01-05 118.0

===== MODE =====
City      Date      AQI_Bucket
0 Ahmedabad 2015-01-01 Moderate
1 Ahmedabad 2015-01-02 Moderate
2 Ahmedabad 2015-01-03 Moderate
3 Ahmedabad 2015-01-04 Moderate
4 Ahmedabad 2015-01-05 Moderate

===== FORWARD FILL =====
City      Date      PM2.5  PM10    NO      NO2      NOx      NH3      CO      SO2 \
0 Ahmedabad 2015-01-01 NaN      NaN      0.92    18.22    17.15    NaN      0.92    27.64
1 Ahmedabad 2015-01-02 NaN      NaN      0.97    15.69    16.46    NaN      0.97    24.55
2 Ahmedabad 2015-01-03 NaN      NaN      17.40   19.30    29.70    NaN      17.40   29.07
3 Ahmedabad 2015-01-04 NaN      NaN      1.70    18.48    17.97    NaN      1.70    18.59
4 Ahmedabad 2015-01-05 NaN      NaN      22.10   21.42    37.76    NaN      22.10   39.33

      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0 133.36    0.00    0.02    0.00  NaN      NaN
1  34.06    3.68    5.50    3.77  NaN      NaN
2  30.70    6.80   16.40    2.25  NaN      NaN
3  36.08    4.43   10.14    1.00  NaN      NaN
4  39.31    7.01   18.89    2.78  NaN      NaN

===== BACKWARD FILL =====
City      Date      PM2.5  PM10    NO      NO2      NOx      NH3      CO \
0 Ahmedabad 2015-01-01 73.24   141.54  0.92    18.22    17.15    26.64  0.92
1 Ahmedabad 2015-01-02 73.24   141.54  0.97    15.69    16.46    26.64  0.97
2 Ahmedabad 2015-01-03 73.24   141.54  17.40   19.30    29.70    26.64  17.40
3 Ahmedabad 2015-01-04 73.24   141.54  1.70    18.48    17.97    26.64  1.70
4 Ahmedabad 2015-01-05 73.24   141.54  22.10   21.42    37.76    26.64  22.10

      SO2      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0 27.64 133.36    0.00    0.02    0.00 209.0  Poor
1 24.55  34.06    3.68    5.50    3.77 209.0  Poor
2 29.07  30.70    6.80   16.40    2.25 209.0  Poor

      SO2      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
28 49.52  59.76    0.02    0.00    3.14 209.0  Poor
29 48.49  97.07    0.04    0.00    4.81 328.0  Very Poor
30 67.39 111.33    0.24    0.01    7.67 514.0  Severe
31 75.23 102.70    0.40    0.04   25.87 782.0  Severe
32 55.04 107.38    0.46    0.06   35.61 914.0  Severe

===== FINAL CLEANED DATASET =====
City      Date      PM2.5  PM10    NO      NO2      NOx      NH3      CO \
0 Ahmedabad 2015-01-01 48.57   95.68  0.92    18.22    17.15   15.85  0.92
1 Ahmedabad 2015-01-02 48.57   95.68  0.97    15.69    16.46   15.85  0.97
2 Ahmedabad 2015-01-03 48.57   95.68  17.40   19.30    29.70   15.85  17.40
3 Ahmedabad 2015-01-04 48.57   95.68  1.70    18.48    17.97   15.85  1.70
4 Ahmedabad 2015-01-05 48.57   95.68  22.10   21.42    37.76   15.85  22.10

      SO2      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0 27.64 133.36    0.00    0.02    0.00 118.0  Moderate
1 24.55  34.06    3.68    5.50    3.77 118.0  Moderate
2 29.07  30.70    6.80   16.40    2.25 118.0  Moderate
3 18.59  36.08    4.43   10.14    1.00 118.0  Moderate
4 39.33  39.31    7.01   18.89    2.78 118.0  Moderate

===== MISSING VALUES AFTER CLEANING =====
City      0
Date      0
PM2.5     0
PM10      0
NO         0
NO2        0
NOx        0
NH3        0
CO         0
SO2        0
O3         0
Benzene    0
Toluene    0
Xylene     0
AQI        0
AQI_Bucket 0
dtype: int64

Data cleaning completed successfully.

```

Result: